

ELEC4631 Solutions Tutorial 9

1. Using the basis decomposition $Y = y_{11} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + y_{21} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + y_{22} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ (Y is symmetric) and $X = x_{11} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + x_{21} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + x_{12} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + x_{22} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ (X is not symmetric), one can write

$$L(X, Y) = \begin{bmatrix} -1 & 3 \\ 3 & -4 \end{bmatrix} + x_{11} \begin{bmatrix} 6 & 1 \\ 1 & 0 \end{bmatrix} + x_{12} \begin{bmatrix} 0 & 3 \\ 3 & 2 \end{bmatrix} + x_{21} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + x_{22} \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \\ + y_{11} \begin{bmatrix} 6 & -10 \\ -10 & 6 \end{bmatrix} + y_{21} \begin{bmatrix} -18 & 14 \\ 14 & 46 \end{bmatrix} + y_{22} \begin{bmatrix} 12 & 6 \\ 6 & -72 \end{bmatrix},$$

which is an LMI in standard form with decision variable $x = (x_{11}, x_{21}, x_{12}, x_{22}, y_{11}, y_{21}, y_{22})$.

2. (a) The LMI from Problem 4 of Tutorial 8 is

$$F(x) = \begin{bmatrix} -3x_1 + x_2 & -x_1 + x_2 - 1 \\ -x_1 + x_2 - 1 & x_1 + x_2 \end{bmatrix} > 0.$$

By the Schur complement, the LMI is equivalent to $x_1 + x_2 > 0$ and $x_2 > 2x_1^2 + x_1 + 1/2$. One can then sketch the region in $\mathbb{R}^2$ defined by these two inequalities as shown in the tutorial. In fact, it turns out that the line $x_1 + x_2 = 0$ just touches (i.e., is tangent to) the parabola $x_2 = 2x_1^2 + x_1 + 1/2$ rather than intersecting. From this sketch one concludes that the solution set to the LMI is $\{(x_1, x_2) \mid x_2 > 2x_1^2 + x_1 + 1/2\}$, which is a region just above the parabola (but does not include the parabola due to the > 0 , rather than ≥ 0). It can be seen from the sketch that the set is convex. This solution set is identical to the one that was obtained for Problem 4 of Tutorial 8.

- (b) By the Schur complement, the LMI is equivalent to $x_1 + x_2 > 0$ and $(x_2 + \frac{5}{2}x_1)^2 - \frac{13}{4}x_1^2 = (x_2 + (\frac{5}{2} - \frac{\sqrt{13}}{2})x_1)(x_2 + (\frac{5}{2} + \frac{\sqrt{13}}{2})x_1) > 0$. Define $S_1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 + x_2 > 0\}$, $S_2 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 > (-\frac{5}{2} + \frac{\sqrt{13}}{2})x_1 \text{ and } x_2 > (-\frac{5}{2} - \frac{\sqrt{13}}{2})x_1\}$, and $S_3 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 < (-\frac{5}{2} + \frac{\sqrt{13}}{2})x_1 \text{ and } x_2 < (-\frac{5}{2} - \frac{\sqrt{13}}{2})x_1\}$. The solution set of the LMI is $S = (S_1 \cap S_2) \cup (S_1 \cap S_3)$, sketch $S_1 \cap S_2$ and $S_1 \cap S_3$,

this is shown in the figure on the next page. From the sketch, it turns out that $S_1 \cap S_3$ is an empty set (S_1 does not intersect S_3), while $S_1 \cap S_2$ is not empty. Therefore the solution set of the LMI is simply $S = (S_1 \cap S_2) \cup \phi = S_1 \cap S_2$, and the sketch shows that this set is convex.

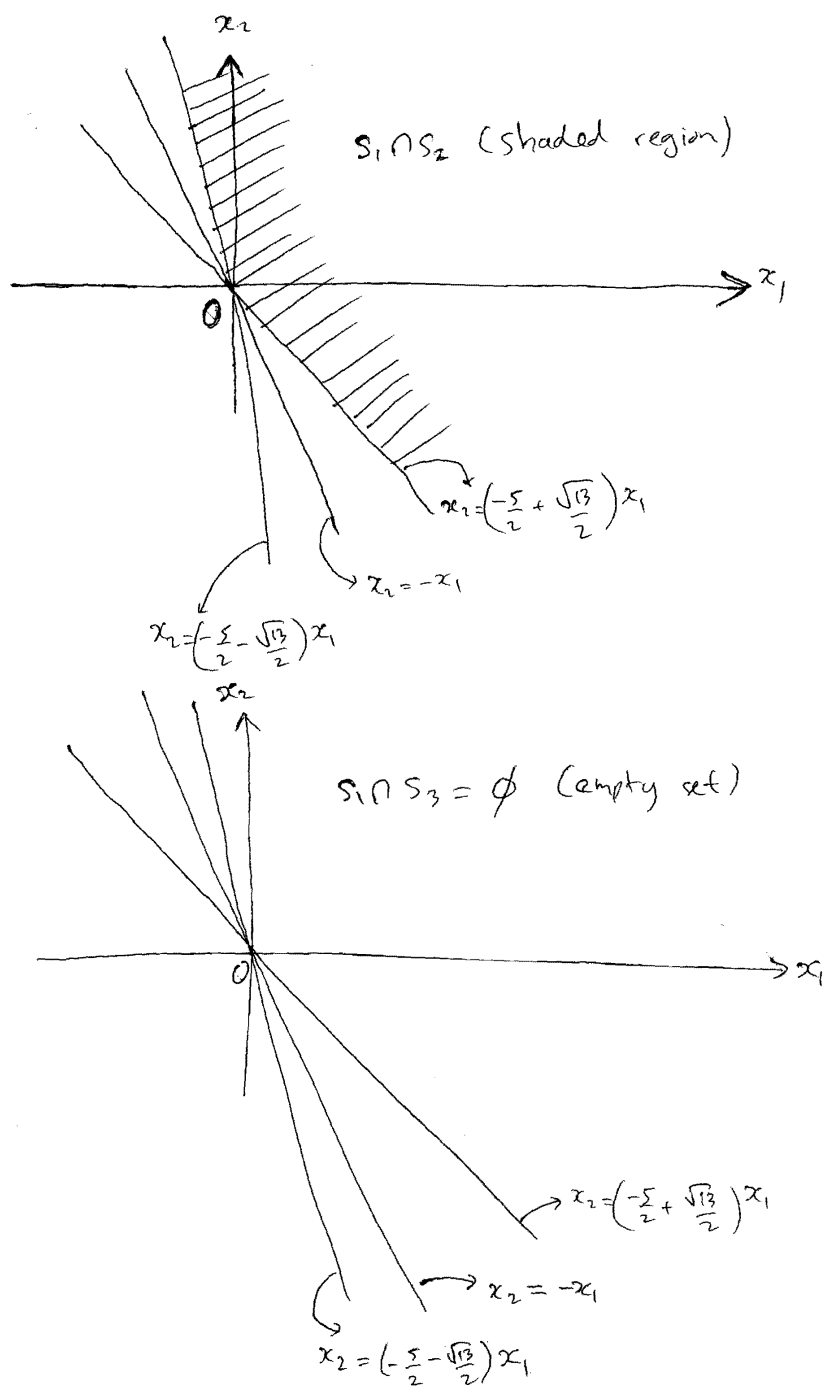


Figure 1: Sketch for Q2 (b).

3. We can write the inequalities

$$\begin{aligned} X &> 0, \\ (A - BK)X(A - BK)^T - X &< 0, \end{aligned}$$

equivalently (using the fact that X is symmetric) as

$$\begin{aligned} X &> 0, \\ X - (A - BK)XX^{-1}X^T(A - BK)^T &> 0. \end{aligned}$$

By the Schur complement, the last inequalities are equivalent to

$$\begin{bmatrix} X & (A - BK)X \\ X^T(A - BK)^T & X \end{bmatrix} > 0.$$

However, the above is still not an LMI since the two off diagonal blocks (that are transposes of one another) are not affine maps, since there are product terms containing both matrix variables X and K . To transform to an LMI introduce the new variable $Y = KX$. Then the above becomes the following LMI in the matrix variables $X = X^T$ and Y :

$$\begin{bmatrix} X & AX - BY \\ XA^T - Y^TB^T & X \end{bmatrix} > 0.$$

If the above LMI is strictly feasible with a solution X and Y then K can be recovered as $K = YX^{-1}$.

4. Since the system has a convergence rate larger than μ , all the eigenvalues of A have eigenvalues with real part less than $-\mu$. We first show that $A + \mu I$ has eigenvalues on the left half plane. Decompose A as $A = VJ_A V^{-1}$ with J_A a Jordan canonical form of A . Since $A + \mu I = V(J_A + \mu I)V^{-1}$, $A + \mu I$ is similar to $J_A + \mu I$. Therefore, $c_{A+\mu I}(s) = c_{J_A+\mu I}(s)$. $J_A + \mu I$ is also in Jordan canonical form, it is an upper triangular matrix and has its eigenvalues on its diagonal. Thus all the eigenvalues of $J_A + \mu I$ are of the form $\lambda + \mu$ with λ an eigenvalue of A . Since all the eigenvalues of A have real part less than $-\mu$, it follows that all the eigenvalues of $J_A + \mu I$ are all in the LHP and therefore also all the eigenvalues of $A + \mu I$.

From Lecture 12, it follows that the LMI

$$\begin{aligned} P &> 0 \\ (A + \mu I)^T P + P(A + \mu I) &< 0 \Leftrightarrow A^T P + P A < -2\mu P \end{aligned}$$

is feasible. This is equivalent to the LMI

$$\begin{aligned} P &> 0 \\ A^T P + P A &\leq -2\mu P \end{aligned}$$

being strictly feasible.

5. (a) The feedback design LMI is (X_f is symmetric and Y_f is not)

$$\begin{aligned} X_f &> 0, \\ AX_f + X_f A^T - BY_f - Y_f^T B^T &< -2\mu_{s,\min} X, \\ AX_f + X_f A^T - BY_f - Y_f^T B^T &> -2\mu_{s,\max} X. \end{aligned}$$

If the above LMI is feasible then $K = Y_f X_f^{-1}$.

The observer design LMI is (X_o is symmetric and Y_o is not)

$$\begin{aligned} X_o &> 0, \\ X_o A + A^T X_o - C^T Y_o - Y_o^T C &< -2\mu_{o,\min} X, \\ X_o A + A^T X_o - C^T Y_o - Y_o^T C &> -2\mu_{o,\max} X. \end{aligned}$$

If the above LMI is feasible then $L = X_o^{-1} Y_o^T$.

- (b) By Matlab, it turns out both the LMIs above are feasible with

$$X_f = \begin{bmatrix} 0.0150 & -1.7865 \\ -1.7865 & 447.1594 \end{bmatrix}; Y_f = 10^3 \begin{bmatrix} -0.0018 & 1.1072 \end{bmatrix}; K = \begin{bmatrix} 337.6963 & 3.8253 \end{bmatrix}.$$

and

$$X_o = \begin{bmatrix} 0.0567 & 4.8727 \\ 4.8727 & 425.2246 \end{bmatrix}; Y_o = 10^3 \begin{bmatrix} 0.0197 & 1.9089 \end{bmatrix}; L = 10^3 \begin{bmatrix} -2.3795 \\ 0.0318 \end{bmatrix}.$$

Checking, the eigenvalues of $A - BK$ are $-1.4177 \pm 1.1693j$, so indeed $1 < \mu_s < 2$. On the other hand eigenvalues of $A - LC$ is $-3.4855 \pm 3.3780j$, so also $3 < \mu_o < 4$ is satisfied.